Field Hours

CSCE 653 Computer Methods in Applied Sciences

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Absolute error and truncation

You report absolute error and relative error as different divisions. Absolute error is the absolute value of approximate minus true. Relative error divides that by the absolute value of the true value. Approximate 1/3 by 0.3, that is 3/10. The difference 3/10 - 1/3 has absolute value 1/30. Divide by 1/3 and the relative error is 1/10. An absolute miss of 1/30 is a different story when the true value is near 1000. You state the scale.

You separate truncation from rounding. The series for e to the x starts 1 + x + x squared over 2. Keep only 1 + x. At x = 0.1 the first dropped term is 0.01 / 2 = 0.005. That is truncation. Rounding is storing 1/3 as 0.3, which made the absolute error 1/30. You can round little and still truncate badly. Name the ledger before you compare methods.

Two errors for one third

Absolute error

3/10 minus 1/3

Absolute value 1/30

Relative error

Divide that by 1/3

Relative error is 1/10

Missing one third by 0.3

1. The difference 3/10 - 1/3 equals -1/30.

2. The absolute error is 1/30. The relative error is (1/30) divided by (1/3), which is 1/10.

3. At x = 0.1, cutting the exponential series after 1 + x drops a first term of size 0.005. That ledger is truncation.

Problem 1. Approximate 1/3 by 33/100. What is the absolute error as a fraction?

Answer. The absolute value of 33/100 - 1/3 is 1/300. That is smaller than 1/30, the error of 3/10.

Problem 2. You keep terms through x squared over 2 for e to the x at x = 0.1, and you drop x cubed over 6. What is the size of that first dropped term?

Answer. x cubed is 1/1000, divided by 6 is 1/6000. Keeping the quadratic term replaced a drop of 0.005 by a drop of 1/6000.

Lecture 2 · 90 minutes

Bisection, then a tangent

You bracket a root by bisection when a continuous function changes sign. For f(x) = x squared - 2 on [1, 2], f(1) = -1 and f(2) = 2. The midpoint 1.5 has f(1.5) = 0.25, so the root lies in [1, 1.5]. One halving cuts the width from 1 to 1/2. After k steps the width is 1 over 2 to the k. You always keep a bracket. You do not need the derivative for this method.

You drop the bracket when you use Newton's update, x minus f(x) divided by f'(x). Here the derivative is 2x. From x = 1 the next value is 1 - (1 - 2) / 2 = 1.5. One more step is 1.5 - (2.25 - 2) / 3 = 1.5 - 1/12 = 17/12. You followed the tangent. If the derivative at your start is 0, the division is undefined and you do not take the step.

Two Newton steps from 1

1. Start at x = 1

2. First update is 1.5

3. Second update is 17/12

From 1.5 down to 17/12

1. f(1.5) = 2.25 - 2 = 0.25, and the derivative at 1.5 is 3.

2. Subtract 0.25 / 3 = 1/12 from 1.5.

3. 1.5 is 18/12, so 18/12 - 1/12 = 17/12. That is the second Newton approximation.

Problem 1. Continue bisection from [1, 1.5]. The midpoint is 1.25 and f(1.25) is negative. What is the next bracket, and what is its width?

Answer. The sign change is between 1.25 and 1.5, so the bracket is [1.25, 1.5]. The width is 0.25, which is 1/4 after two halvings of the original width 1.

Problem 2. Take one Newton step on the same f starting at x = 2. What value do you get?

Answer. f(2) = 2 and f'(2) = 4, so you compute 2 - 2/4 = 1.5.

Lecture 3 · 90 minutes

A line, then a square

You get the line through two samples from the slope. Points (0, 1) and (2, 5) have slope (5 - 1)/(2 - 0) = 2, so y = 1 + 2x. At x = 1, with no sample, you predict 3. Two distinct x values fix degree 1. A third sample generally forces a quadratic. You drop the line when the third point misses it.

You can watch that line miss a square. Samples of x squared at 0 and 2 are (0, 0) and (2, 4), so the line is y = 2x. At x = 1 the line says 2 and the function says 1. The absolute error is 1. Add (1, 1) and the quadratic is y = x squared, with error 0 at x = 1. Matching degree removes this polynomial error. It does not repair a bad sample.

Line against the square

Line on the ends

Predicts 2 at x = 1

True value is 1

Quadratic, three points

Hits the middle sample

Error at x = 1 is 0

Error 1 on the chord

1. Samples of x squared at 0 and 2 are 0 and 4. The slope is 2, and the line is y = 2x.

2. At x = 1 the line returns 2 and the function returns 1. Absolute error 1.

3. The extra sample (1, 1) makes the interpolant y = x squared, so the error at x = 1 drops to 0.

Problem 1. The line through (0, 1) and (2, 5) is y = 1 + 2x. What does it return at x = 2 and at x = 3?

Answer. At x = 2 it returns 5, a sample you already have. At x = 3 it returns 7. The value 7 is past the last sample.

Problem 2. You interpolate x squared using only x = 0 and x = 2, then you evaluate at x = 1. What is the absolute error?

Answer. The line gives 2 and the function gives 1, so the absolute error is 1.

Lecture 4 · 90 minutes

Trapezoid, then Simpson

You estimate an integral with one trapezoid: (b - a) / 2 times (f(a) + f(b)). Integrate x squared from 0 to 2. The antiderivative x cubed over 3 gives 8/3. The trapezoid sees f(0) = 0 and f(2) = 4, so the estimate is 4. The absolute error is 4 - 8/3 = 4/3. A single trapezoid on a bending graph misses by a visible amount. Here the miss is exactly 4/3, not a rounding artifact.

You add the midpoint with Simpson: (b - a)/6 times (f(a) + 4 f(mid) + f(b)). On [0, 2] the mid value is f(1) = 1. Then 2/6 times (0 + 4 + 4) equals 8/3, and the error is 0. Simpson is exact on quadratics, and x squared is one, so the zero is by design. You now have 4 from the trapezoid and 8/3 from Simpson.

Same integral, two rules

Trapezoid

Estimate equals 4

Error equals 4/3

Simpson

Estimate equals 8/3

Error equals 0

x squared from 0 to 2

1. The true integral is 8/3.

2. The trapezoid returns 4, so the absolute error is 4 - 8/3 = 4/3.

3. Simpson returns (1/3) times 8 = 8/3. The error is 0 because the integrand is a quadratic.

Problem 1. Use the trapezoid rule on x squared from 0 to 1. What are the estimate, the true integral, and the absolute error?

Answer. The estimate is 1/2. The true integral is 1/3. The absolute error is 1/2 - 1/3 = 1/6.

Problem 2. Why is Simpson's error 0 on [0, 2] for x squared, while the trapezoid error is not?

Answer. Simpson is exact for quadratics, and x squared has degree 2. The trapezoid is exact for linear functions. Its error here is 4/3, not 0.

Lecture 5 · 90 minutes

Elimination and a swap

You clear a 2 by 2 system by elimination, then back-substitute. Take x + y = 3 and 2x + y = 4. Subtract 2 times the first row from the second. You get -y = -2, so y = 2, then x + 2 = 3, so x = 1. One multiplier, the number 2, did the elimination. Both checks pass: 1 + 2 = 3 and 2 + 2 = 4.

You swap when the pivot has smaller absolute value than the entry below it. The pivot 1 sits above 2, so partial pivoting swaps. The multiplier becomes 1/2, not 2. You still finish at y = 2 and x = 1. On this exact system the swap changes the arithmetic, not the answer. You swap so you do not divide by a smaller pivot.

Clear y, then find x

1. Subtract 2 times row 1

2. Read y = 2

3. Back-substitute x = 1

4. Check both original rows

One multiplier, two checks

1. Rows: x + y = 3 and 2x + y = 4. The multiplier is 2.

2. The second row minus twice the first is -y = -2, so y = 2.

3. Then x + 2 = 3, so x = 1. The checks are 3 and 4, both matching.

Problem 1. Solve x + y = 5 and 3x + y = 7 by elimination without a swap. What are x and y?

Answer. Subtract 3 times the first row from the second to get -2y = -8, so y = 4. Then x + 4 = 5, so x = 1.

Problem 2. In the lecture system, why does partial pivoting refuse the multiplier 2?

Answer. The pivot 1 has smaller absolute value than the 2 below it. You swap, and the multiplier becomes 1/2. The solution remains x = 1, y = 2.

Lecture 6 · 90 minutes

A cubic that does not corner

You use one cubic on each interval, matching value, slope, and second derivative at each interior knot. A natural spline sets the second derivative to 0 at both ends. Samples (0, 0), (1, 1), and (2, 0) make two intervals of width 1. The polyline through them is a linear spline with a corner at x = 1. The cubic is not allowed that corner.

You check the piece 1.5 x - 0.5 x cubed. It is 0 at x = 0 and 1 at x = 1. Its second derivative -3x is 0 at x = 0, the natural end. At x = 0.5 it equals 0.75 - 0.0625 = 11/16. The linear spline returns 1/2, which is 8/16. The gap is 3/16. Matching the slope at the knot lifts the middle above the chord.

Values on the first cubic

x = 0 gives value 0

x = 0.5 gives 11/16

x = 1 gives value 1

Chord versus 11/16

1. The cubic 1.5 x - 0.5 x cubed equals 1 at x = 1, the middle sample.

2. At x = 0.5 it equals 0.75 - 0.0625 = 11/16.

3. The straight segment from (0, 0) to (1, 1) equals 1/2 there. The cubic sits 3/16 higher.

Problem 1. Evaluate 1.5 x - 0.5 x cubed at x = 1. Which sample does that check?

Answer. 1.5 - 0.5 = 1, the sample (1, 1). The spline has to hit every sample, so this check must pass.

Problem 2. What does the linear spline return at x = 0.5, and what is the absolute gap versus 11/16?

Answer. It returns 1/2, which is 8/16. The gap is 11/16 - 8/16 = 3/16.

Lecture 7 · 90 minutes

Euler steps and a second difference

You step y' = y, y(0) = 1, by Euler: add h times the current y. With h = 1/2 the first step is 1 + 1/2 = 3/2. The second is 3/2 + (1/2)*(3/2) = 9/4, now at t = 1. Each step multiplies by 3/2, so two steps multiply by 9/4. You replaced the solution curve by 2 straight segments. That replacement is the truncation.

You approximate u" by (u left - 2u + u right) over h squared. Take h = 1/2 and -u" = 2, with u(0) = u(1) = 0. The true solution x(1 - x) equals 1/4 at x = 1/2. One interior node gives (0 - 2u)/(1/4) = -2, so -8u = -2 and u = 1/4. The node matches because this u is a quadratic and the second difference is exact on it.

Two Euler steps to t = 1

Start at y = 1 -> First step reaches 3/2 -> Second step reaches 9/4 -> Stop at time t = 1

Multiply by 3/2 twice

1. With h = 1/2, the first step is 1 + (1/2) times 1 = 3/2.

2. The second step is (3/2) times (3/2) = 9/4.

3. Separately, the difference equation for -u'' = 2 gives u(1/2) = 1/4, matching x(1 - x) at that node.

Problem 1. Take one Euler step of size h = 1 for y' = y, y(0) = 1. What do you get at t = 1, and how does it differ from 9/4?

Answer. You get 1 + 1 = 2. Two steps of size 1/2 gave 9/4. Both approximate y(1), and 2 is not 9/4.

Problem 2. The second-difference equation produced -8u = -2. What is u?

Answer. u = 1/4. That equals x(1 - x) at x = 1/2.

Lecture 8 · 90 minutes

What you can now check

You can now separate an absolute error of 1/30 from a truncation of size 0.005, bracket a root, and follow Newton's tangent. Two updates of x squared - 2, starting at 1, produced 17/12. The residual is 289/144 - 2 = 1/144. The derivative at 17/12 is 17/6. Divide the residual by that derivative: (1/144) times (6/17) = 1/408. That quotient estimates the absolute error. You still have the bracket method when the derivative vanishes.

You keep integral error and differential error apart. Simpson on x squared from 0 to 2 returned 8/3 with error 0, exact on quadratics. Euler on y' = y, two steps of size 1/2, returned 9/4 at t = 1, which is not exact. You say which polynomial the rule matches. One of these errors is 0. The Newton estimate on a different problem was 1/408.

Residual versus exact integral

Newton at 17/12

Residual is 1/144

Estimate of error 1/408

Simpson on [0, 2]

Estimate is 8/3

Error on this quadratic is 0

Divide 1/144 by 17/6

1. f(17/12) = 289/144 - 288/144 = 1/144. The derivative is 17/6.

2. Divide: (1/144) times (6/17) = 1/408. That estimates the absolute distance to the root.

3. Simpson's 8/3 needs no correction on x squared. Its error is 0, while 1/408 belongs to the root problem.

Problem 1. If the residual stayed 1/144 but the derivative were 6, what absolute-error estimate would the division give?

Answer. (1/144) / 6 = 1/864. You divide the residual by the derivative. A different derivative changes the estimate.

Problem 2. Combine Euler and Simpson. Euler produced 9/4 at t = 1. Simpson produced 8/3 for the integral of x squared from 0 to 2. Which result is exact for its own problem, and what is 8/3 minus 9/4?

Answer. Simpson's 8/3 is exact for that integral. Euler's 9/4 is not exact for the exponential. The difference is 32/12 - 27/12 = 5/12, a comparison of two outputs, not an error term for either problem.